I. SYSTEM INITIALIZATION & SETUP ROUTINE

START: Power On / Reset

HAL_Init & SystemClock_Config 64MHz

MX_GPIO_Init & MX_DMA_Init

MX_ADC1/2_Init, MX_TIM1/2/3_Init,
MX_USART1/3_Init

MX_NVIC_Init: Set Priority Interrupt

HAL_ADCEx_Calibration_Start &
FreeRTOS Task Scheduler

II. MAIN LOOP / TASK SCHEDULER

startMediumFrequencyTask & MC_Scheduler

Check State & Motor Reference Input

State RUN &
Referensi Siap?

TIDAK

PWM Tetap Tidak
Diaktifkan

YA

Entry Cepat PWM/ADC Task

III. HIGH-FREQUENCY ISR / FOC

TSK_HighFrequencyTask / FOC_CurrControllerM1

Baca Arus Fase ADC Injected & Hall Feedback

Clarke Transformation: I_alpha, I_beta

Park Transformation: Id, Iq

PI Current Controller & Limit Voltage

Inverse Park & SVPWM Duty Cycle

TIM1_CCR1/2/3 & TIM1_BDTR MOE Output PWM

IV. FAULT HANDLING & SAFETY ROUTINE

StartSafetyTask: Check Temp, Current, Vbus

Fault Terdeteksi?

YA

TSK_SafetyTask:
PWMOFF / Motor Stop

HALT / SAFE STATE

TIDAK

Permintaan
Shutdown /
Power Off?

YA

power_control:
Nonaktifkan Output

END: Power Off

TIDAK